Supplementary Material

Porous polyimide/hexagonal boron nitride films by non-solvent induced phase separation: a three-phase Lewis–Nielsen description of dielectric and thermal properties

Author names — to be completed

S1 Supplementary model figures

S1.1 Shape-factor sensitivity analysis

Figure S1 examines the sensitivity of the three-phase Lewis–Nielsen predictions to the h-BN shape factor A over the range 0.3–6.0 (needle-like rods to aligned platelets), evaluated at the NP mean porosity of 65.6% with $\phi_{\max} = 0.60$ fixed. For the dielectric model (Figure S1a), all A curves are nearly indistinguishable, confirming that ε_r is dominated by the pore term and that the fitted dielectric A_{BN} is weakly determined. For the thermal model (Figure S1b), all $A = 0.3$–6.0 curves remain below the experimental NP 70 wt% value (0.16–0.18 vs. 0.305 W m^{-1} K^{-1}), reinforcing that the 70 wt% enhancement is driven by proximity to maximum packing and incipient particle contact rather than by any plausible single-particle shape factor. Full discussion appears in Sections 3.2 and 4.6 of the main text.

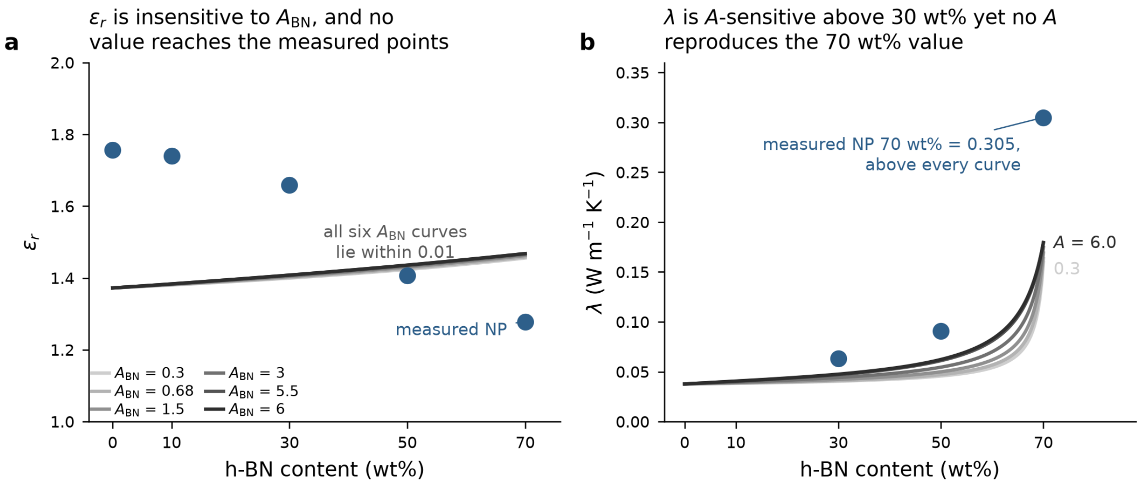

Figure S1: Sensitivity of three-phase LN predictions to h-BN shape factor A ($A = 0.3$–6.0) for (a) dielectric constant and (b) thermal conductivity, evaluated at the NP mean porosity of 65.6% ($\phi_{\max} = 0.60$ fixed). Symbols: measured NP data. In (a) all six curves lie within 0.01 of one another, confirming that ε_r is pore-dominated and A_{BN} weakly determined; note also that no value of A brings the curves onto the measured points. In (b) the curves separate above 30 wt% yet all remain far below the measured NP 70 wt% value of 0.305 W m^{-1} K^{-1}, so that enhancement cannot be attributed to any plausible single-particle shape factor.

S1.2 Parity plot of the dielectric model

Figure S2 compares the three-phase LN prediction with the measured dielectric constant for all ten films. The model RMSE (0.43 NP, 0.36 HP) exceeds the standard deviation of the measured data (0.19 NP, 0.18 HP), so the coefficient of determination is negative (Section 4.2 of the main text); the numerical values are listed in Table 2 of the main text.

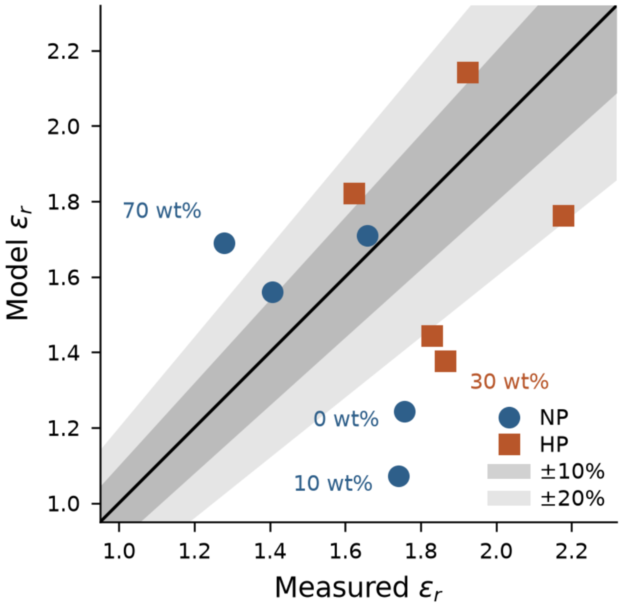

Figure S2: Parity plot of predicted vs. measured dielectric constant for all ten films (three-phase LN model evaluated at each film's open porosity), with ±10% and ±20% bands. Labels identify the compositions with the largest residuals.

S1.3 Porosity sensitivity at fixed composition

Figure S3 shows the calibrated ε_r and λ at fixed 50 wt% h-BN as porosity is varied. Both properties rise as porosity falls, so porosity alone cannot decouple them; the same coupling is shown for every composition by the loci of Figure 4c of the main text.

S1.4 Feasible processing window

Figure S4 shows the porosity range that satisfies each of the three target specifications defined in Section 4.8 of the main text (Basic: $\varepsilon_r < 2.0$, $\lambda > 0.05\,\mathrm{W\,m^{-1}\,K^{-1}}$; Advanced: $\varepsilon_r < 1.8$, $\lambda > 0.08\,\mathrm{W\,m^{-1}\,K^{-1}}$; Future: $\varepsilon_r < 1.6$, $\lambda > 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$), derived from the design maps of Figure 4a,b of the main text.

S1.5 Porosity that maximizes thermal conductivity under a dielectric ceiling

Figure S5 shows, for each h-BN loading, the porosity that maximizes λ under a given ε_r ceiling. The five composition curves stay within 10 percentage points of one another, so the porosity target is set almost entirely by the dielectric ceiling while the achievable λ at that porosity is set by the loading (Section 4.8 of the main text).

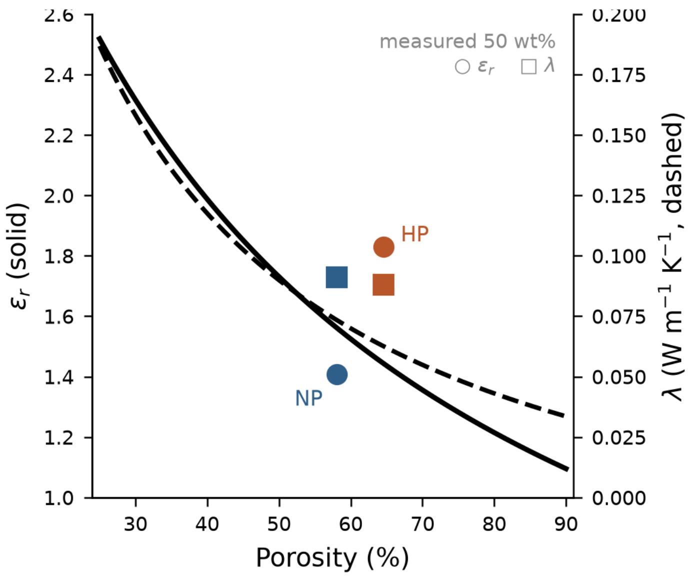

Figure S3: Porosity sensitivity at fixed 50 wt% h-BN: ε_r (left axis, solid) and λ (right axis, dashed) from the calibrated three-phase LN model. Circles and squares mark the measured NP and HP 50 wt% films (ε_r and λ, respectively).

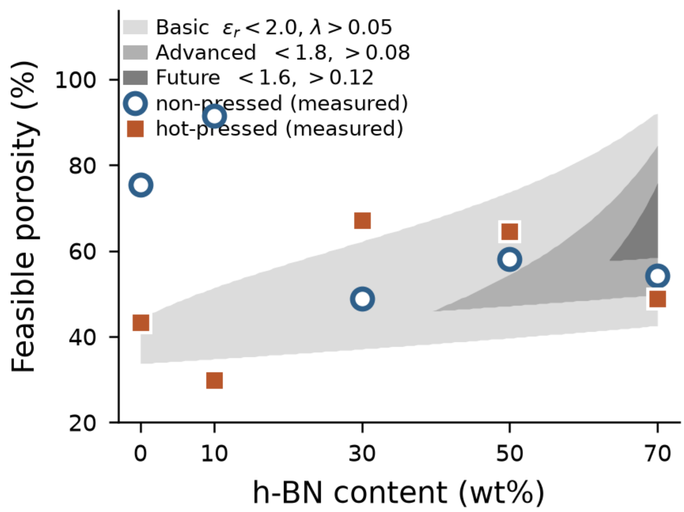

Figure S4: Feasible processing window: the porosity range meeting each target specification vs. h-BN content (nested shading, light to dark for Basic→Future); measured NP (open circles) and HP (filled squares) states overlaid.

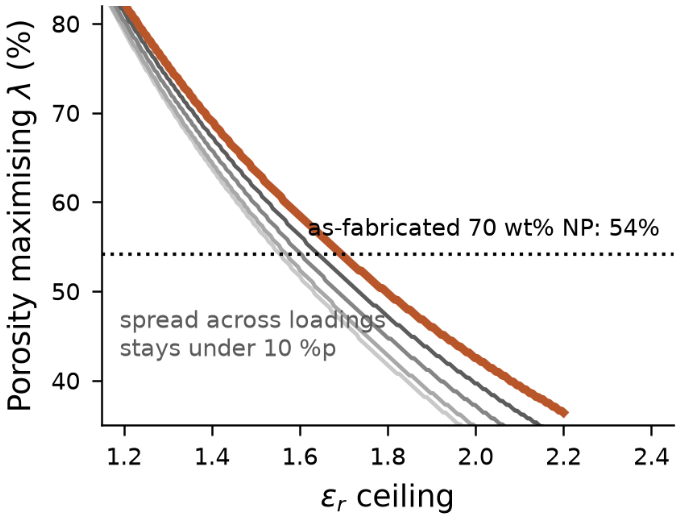

Figure S5: Porosity that maximizes λ under each ε_r ceiling for 0–70 wt% h-BN (light to dark grey, 0–50 wt%; orange, 70 wt%). The dotted line marks the as-fabricated porosity of the 70 wt% NP film (54%).

S2 Additional materials characterization

S2.1 Film photographs

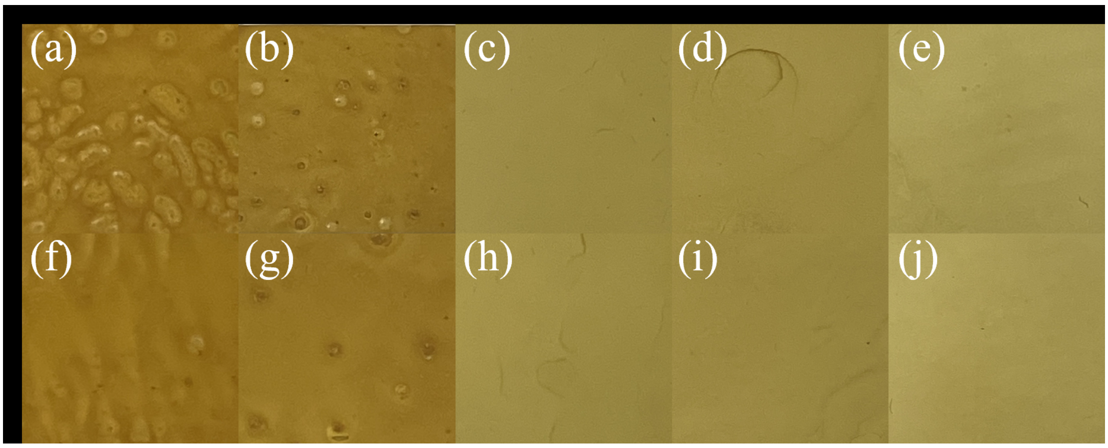

Figure S6: Photographs of the PI/h-BN films: (a–e) non-pressed and (f–j) hot-pressed at 0, 10, 30, 50, and 70 wt% h-BN.

Add composition column headers, NP/HP row labels, and a scale bar directly on the figure, and crop the black margin.

S2.2 Top-view SEM

S2.3 FT-IR

S2.4 EDS elemental analysis

To be added: mercury intrusion pore-size distributions for all ten films. The comparison of open (intrusion) and total (density-derived) porosity is given in Table 3 of the main text.

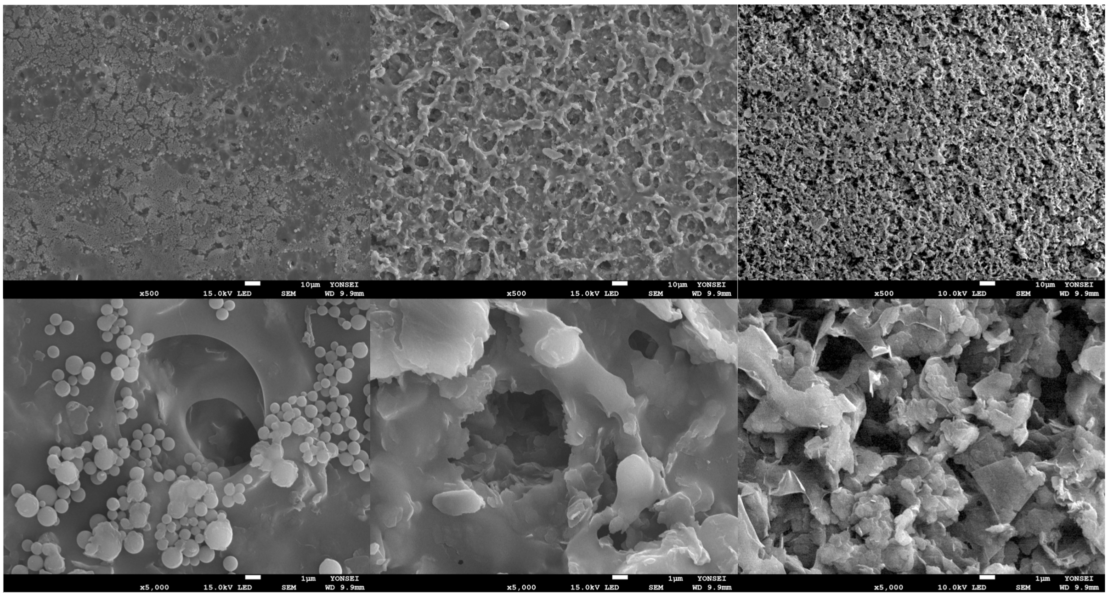

Figure S7: Top-view FE-SEM micrographs of PI/h-BN films at 30, 50, and 70 wt% h-BN (columns) at two magnifications (rows). Label the rows by magnification (×500, ×5,000) and the columns by composition on the figure.

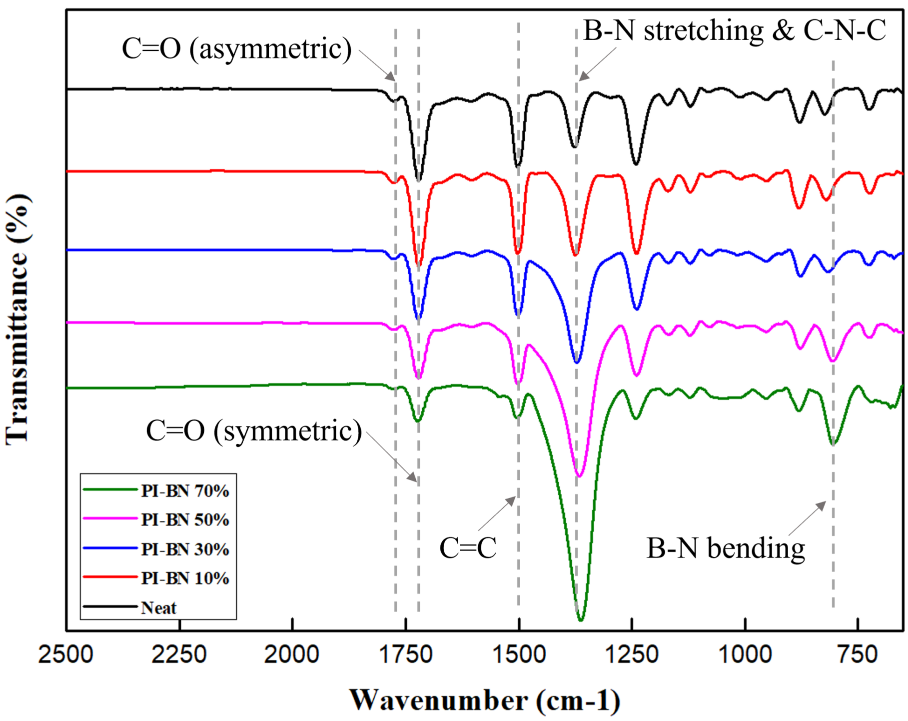

Figure S8: FT-IR spectra (Jasco FT/IR-460 Plus, 600–4000 cm^{-1}) confirming complete imidization. Confirm the band positions against the recorded spectra and use one set of values here and in Section 2.3 of the main text (currently 1720/1780 and 1380 cm^{-1} in the main text vs. 1722/1778 and 1366 cm^{-1} in an earlier draft of this caption). Re-plot the figure with a cm^{-1} axis label and the typeface of the other figures.

Table S1: EDS elemental composition (wt%) of PI/h-BN films (FE-SEM/EDS, JEOL-7800F). B content scales with nominal h-BN loading at ≥10 wt% and is preserved upon hot-pressing.

Sample	B	C	N	O	Total
NP neat	5.41†	73.72	3.83	17.04	100
NP 10 wt%	7.99	65.55	10.42	16.04	100
NP 30 wt%	14.70	54.28	19.36	11.66	100
NP 50 wt%	19.80	44.10	26.49	9.60	100
NP 70 wt%	29.28	26.65	38.62	5.45	100
HP neat	—	75.17	0.00	24.83	100
HP 10 wt%	8.54	68.30	8.67	14.49	100
HP 30 wt%	14.77	54.74	18.85	11.64	100
HP 50 wt%	21.11	43.12	26.62	9.15	100
HP 70 wt%	28.17	29.15	36.48	6.20	100

A second NP 70 wt% spot gave B/C/N/O = 28.81/27.93/37.01/6.25 wt%, confirming spatial uniformity.
†The apparent B signal in the neat NP film, which contains no boron, reflects the EDS quantification limit for light elements (B Kα overlaps the C/N/O region) and is treated as background.

S3 Model details and benchmarks

S3.1 Sequential three-phase Lewis–Nielsen formulation

To be completed: full derivation and parameter definitions (main-text Eqs. 1–3 expanded), including the step-1 (dense composite) and step-2 (pore incorporation) expressions and the packing-interaction term ψ. Number the key expression as Eq. (S8); the main text cites Eq. S8.

S3.2 Volume-fraction conversion

To be completed: wt% → vol% conversion using $\rho_{\mathrm{PI}} = 1.42\,\mathrm{g\,cm^{-3}}$ and $\rho_{\mathrm{BN}} = 2.29\,\mathrm{g\,cm^{-3}}$, with the porosity-corrected film volume fractions used throughout, and the ρ_{solid} values behind the ϕ_{total} column of Table 3 of the main text.

S3.3 Sensitivity of the dielectric fit

To be completed. Include a re-fit of the dielectric model using ϕ_{total} (density-derived) as the porosity input for the seven films with measured density, reporting RMSE and R^2 side by side with the open-porosity fit used in the main text.

S3.4 Uncertainty propagation

To be completed: propagation of the α, C_p and ρ uncertainties into λ (about 10%), and the uncertainty of ε_r including the inter-laboratory comparison of Section 4.2.

S3.5 Benchmark against standard mixing rules

To be completed: quantitative comparison of the calibrated LN model against Lichtenecker, Maxwell–Garnett, and Bruggeman zero-parameter mixing rules (the main text cites RMSE 0.26 vs. 0.55–0.90 over the 30–70 wt% films; state whether NP-only or NP+HP films are included so that the value is consistent with the NP-only RMSE of 0.25 quoted in Section 4.2).

S4 Parameter estimation protocol

S4.1 Dielectric calibration

To be completed: objective function, optimizer settings, initial values, and convergence diagnostics.

S4.2 Degeneracy of the unconstrained thermal fit

To be completed: demonstration that the unconstrained three-parameter thermal fit converges to a degenerate solution ($A \to 700$, $\phi_{\max} \to 0.10$) with correlated parameters, motivating the single-parameter calibration ($\lambda_{\mathrm{BN,eff}}$ fitted; $A = 5.5$, $\phi_{\max} = 0.60$ fixed). The main text cites Section S4.2. Include the RMSE comparison table (constrained 0.027/0.030 vs. unconstrained 0.027/0.023 $\mathrm{W\,m^{-1}\,K^{-1}}$); the main text references Table S11.

S5 Python implementation

To be completed: complete listing of the three-phase Lewis–Nielsen model and parameter-estimation code (the Excel workbook 'Model_Calculator' sheet implements the identical pipeline in spreadsheet form). Deposit the code in a public repository and reference the DOI here and in the Data availability statement.